

Enhanced tunable X-rays from bulk crystals driven by table-top free electron energies

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S1. Intensity of tunable X-rays from free electron-driven crystals

The tunable, narrowband, coherent X-rays that are emitted from free-electron-driven crystals are dominated by two processes, namely parametric X-ray radiation (PXR) and coherent bremsstrahlung (CBS). PXR arises from the diffraction of the incident electron's Coulomb field by the crystal's periodic atomic structure, while CBS is generated by the periodic undulation of electrons by the periodic potential as it traverses the crystal. The theory of tunable X-ray generation is based on the framework proposed by Baryshevsky *et al.*^{1,2} and further extended by later studies to include the relativistic corrections and the electron scattering inside the crystal³⁻⁶. The far-field radiation intensity from a moving charged particle is given by

$$\frac{d^2N}{d\Omega d\omega} = \frac{4\pi c \epsilon_0}{\hbar \omega} R^2 |\mathbf{E}(\mathbf{r}, \omega)|^2, \quad (1)$$

Where N is the number of emitted photons, Ω is the solid angle, ω is the angular frequency of the emitted photon, c is the speed of light in free space, ϵ_0 is the vacuum permittivity, $\hbar$ is the reduced Planck's constant, R is the distance between the charge and the observation point, $\mathbf{r}$ is the position vector in space, $\mathbf{E}(\mathbf{r}, \omega)$ represents the radiation electric field, which can be obtained by solving Maxwell's equations in non-magnetic linear media. By substituting the $\mathbf{E}(\mathbf{r}, \omega)$ in Supplementary Equation (1), the X-ray intensity is

$$\frac{d^2N}{d\omega d\Omega} = \frac{1}{N_e} \frac{\alpha \omega}{4\pi^2 c^2} \sum_{j=1}^{N_e} \sum_{i=1}^{N_{\text{coll}}^j} \left| \int_0^{t_L} \mathbf{v}_i(t) \cdot \mathbf{E}_{\text{ks}}[\mathbf{r}_i(t), \omega] e^{-i\omega t} dt \right|^2, \quad (2)$$

where N is the number of emitted photons, ω is the angular frequency of the emitted photon, Ω is the solid angle, N_e is the number of electrons incident on the crystal, α is the fine-structure constant, t_L is the electron interaction time with the crystal of the i^{th} electron trajectory segment, $\mathbf{v}_i(t)$ is the velocity of the i^{th} electron trajectory segment of the j^{th} electron at time t , $\mathbf{E}_{\text{ks}}[\mathbf{r}_i(t), \omega]$ represents the eigenmode of the crystal and is given by

$$\mathbf{E}_{\mathbf{k}s}[\mathbf{r}(t), \omega] = e^{i\mathbf{k}\cdot\mathbf{r}}\hat{\mathbf{e}}_s + \sum_{\mathbf{g}\neq 0} -\chi_{\mathbf{g}}(\omega) \frac{\left[(\mathbf{k} + \mathbf{g})(\mathbf{g} \cdot \hat{\mathbf{e}}_s) - \frac{\omega^2}{c^2}\hat{\mathbf{e}}_s\right]}{\left(k_{\mathbf{g}}^2 - \frac{\omega^2}{c^2}\right)} e^{i(\mathbf{k}+\mathbf{g})\cdot\mathbf{r}}, \quad (3)$$

where

$$\chi_{\mathbf{g}}(\omega) = -\frac{4\pi e^2}{\epsilon_0 m_e \omega^2} \frac{s(\mathbf{g})}{V} e^{-W(\mathbf{g})}, \quad (4)$$

where $\mathbf{k}$ is the wave vector of the radiation field, s is the variable for the two possible polarizations (s-polarized and p-polarized), $\mathbf{r}(t)$ is the position coordinate of the electron, $\hat{\mathbf{e}}_s$ is the polarization vector, $\mathbf{g}$ is the reciprocal lattice vector of the crystal, $\chi_{\mathbf{g}}(\omega)$ is the Fourier component of electric susceptibility, e is the elementary charge value, $s(\mathbf{g})$ is the crystal structure factor, V is the volume of the unit cell, $e^{-W(\mathbf{g})}$ represents the Debye-Waller factor, which accounts for the effects of thermal vibrations in a crystal⁷.

Assuming the fluctuations in an electron trajectory segment between two consecutive collisions are small, which always hold in our regime of interest, Supplementary Equation (2) can be simplified to

$$\frac{d^2 N}{d\omega d\Omega} \approx \frac{1}{N_e} \frac{\alpha \omega}{4\pi^2 c^2} \sum_{j=1}^{N_e} \sum_{i=1}^{N_{\text{coll}}^j} \left| \sum_{\mathbf{g}\neq 0} A_{\mathbf{g}s}(\omega, \hat{\mathbf{n}}) \right|^2, \quad (5)$$

where

$$\begin{aligned} A_{\mathbf{g}s}(\omega, \hat{\mathbf{n}}) &= A_{\text{PXR}} + A_{\text{CBS}}, \\ A_{\text{PXR}} &= Q(\mathbf{v} \cdot \boldsymbol{\epsilon}_{\mathbf{g}s}), \\ A_{\text{CBS}} &= Q\left(\frac{\xi_{\mathbf{g}}}{2\gamma} [\{\mathbf{g}; \hat{\mathbf{e}}_s\} + \Lambda]\right) + Q'\left(\frac{\xi_{\mathbf{g}}}{2\gamma} [\{\mathbf{g}; \hat{\mathbf{e}}_s\} - \Lambda]\right), \end{aligned} \quad (6)$$

where the notation Q , Q' , $\xi_{\mathbf{g}}$, $\{\mathbf{a}; \mathbf{b}\}$ and Λ denotes

$$Q = \frac{1}{2\mathcal{P}} [\sin 2\mathcal{P}t_L + i(\cos 2\mathcal{P}t_L - 1)], \quad \mathcal{P} = \frac{\omega - \mathbf{v} \cdot (\mathbf{k} + \mathbf{g})}{2}, \quad (7)$$

$$Q' = \frac{1}{2\mathcal{P}'} [\sin 2\mathcal{P}'t_L + i(\cos 2\mathcal{P}'t_L - 1)], \quad \mathcal{P}' = \frac{\omega - \mathbf{v} \cdot (\mathbf{k} - \mathbf{g})}{2},$$

$$\xi_{\mathbf{g}} = -\frac{qU_{\mathbf{g}}}{mV(\mathbf{v} \cdot \mathbf{g})}$$

$$\{\mathbf{a}; \mathbf{b}\} = a_x b_x + a_y b_y + \frac{a_z b_z}{\gamma^2},$$

$$\Lambda = \{\mathbf{k}; \mathbf{g}\} \frac{\mathbf{v} \cdot \hat{\mathbf{e}}_s}{\mathbf{v} \cdot \mathbf{g}}.$$

In the above expressions, $A_{\mathbf{g}s}$ is the radiation amplitude, A_{PXR} and A_{CBS} are radiation amplitude for PXR and CBS respectively, $\hat{\mathbf{n}}$ is the unit vector along the observation direction, γ is the Lorentz factor, t_L is the electron interaction time with the crystal of the i^{th} electron trajectory segment, where L is the interaction length of the i^{th} electron trajectory segment, which is obtained by using Monte Carlo simulation of electron scattering, and $\mathbf{v}$ is the electron velocity, q is the charge of the incident particle and $U_{\mathbf{g}}$ is a constant related to the crystal atomic potential $U(\mathbf{r})$, given as

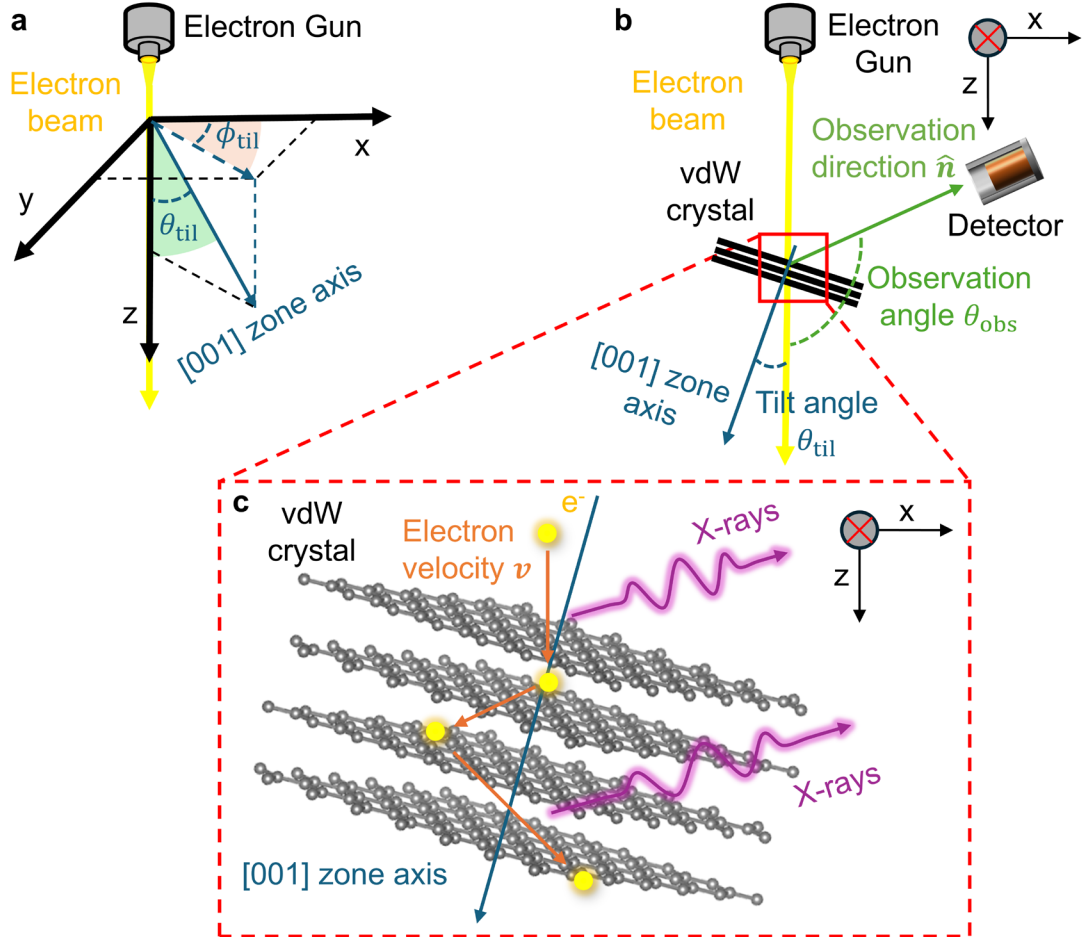
$$U(\mathbf{r}) = \frac{1}{2V} \sum_{\mathbf{g} \neq 0} U_{\mathbf{g}} e^{i\mathbf{g} \cdot \mathbf{r}} + c. c., \quad (8)$$

$$U_{\mathbf{g}} = \frac{e}{\epsilon_0} \sum_i e^{i\mathbf{g} \cdot \mathbf{R}_i} \left[\frac{Z_i - F_i(\mathbf{g})}{g^2} \right] e^{-W(\mathbf{g})},$$

where Z_i is the i^{th} atom's nuclear charge in the unit cell, $F_i(\mathbf{g})$ is the form factor of the i^{th} atom in the unit cell and $\mathbf{R}_i$ is the coordinate of the i^{th} atom in the unit cell, $c. c.$ denotes complex conjugate. Supplementary Equations (6) and (7) imply the X-ray intensity has its peak value when $\mathcal{P} \rightarrow 0$ and $\mathcal{P}' \rightarrow 0$, resulting in $|Q'|^2 \rightarrow t_L^2$ and $|Q|^2 \rightarrow t_L^2$. In our regime of interest, the X-ray peak is dominated by parametric X-ray radiation. The photon energy at the peak radiation intensity (when $\mathcal{P} \rightarrow 0$) is obtained as

$$E_p = \hbar c \frac{\mathbf{v} \cdot \mathbf{g}}{c - \mathbf{v} \cdot \hat{\mathbf{n}}}, \quad (9)$$

where $\hbar$ is the reduced Planck constant, and $\mathbf{v}$ is the electron velocity, $\mathbf{g}$ is the reciprocal lattice vector of a certain plane of the crystal, $\hat{\mathbf{n}}$ is the unit vector along the observation direction, c is the speed of light in vacuum. The geometrical relationship is further illustrated by Supplementary Fig. 1.



Supplementary Fig. 1. Schematic representation of the X-ray generation process from free-electron-driven crystals (here, vdW crystals), tilt and observation angles, and the coordinate system. **a** The coordinate system is defined with the z-axis aligned with the initial electron beam direction. The tilt of the crystal's [001] zone axis is specified by the polar angle θ_{til} and azimuthal angle ϕ_{til} . **b** Schematic of the electron beam interacting with the crystal to generate X-rays. The polar angle θ_{obs} defines the direction of the observation vector $\hat{\mathbf{n}}$, measured as the angle between the observation vector and the initial electron beam direction (aligned with the z-axis). **c** Illustration of electron scattering within the van der Waals (vdW) material's crystal lattice and the corresponding narrowband, coherent X-ray generation process.

S2. Monte Carlo simulation of electron scattering

The electron scattering process described in the theoretical framework above is accounted for using Monte Carlo simulations conducted with the CASINO software⁸⁻¹², developed in C++. The initial landing position of an electron on the sample is assumed to follow a Gaussian distribution

$$\begin{aligned} X_0 &= \frac{d\sqrt{\log R_1}}{2 \times 1.65} \cos(2\pi R_2), \\ Y_0 &= \frac{d\sqrt{\log R_1}}{2 \times 1.65} \cos(2\pi R_3), \end{aligned} \quad (10)$$

where X_0 , and Y_0 are the electron coordinates, d is the diameter of the electron beam spot containing 99.9% of the total electron distribution, R_i are random numbers uniformly distributed between 0 and 1. The distance L between two successive collisions is determined as

$$\begin{aligned} L &= -\lambda_{el} \log R_4, \\ \frac{1}{\lambda_{el}} &= \rho N_0 \sum_{i=1}^n \frac{C_i \sigma_{el}^i}{A_i}, \end{aligned} \quad (11)$$

where L is the distance in nanometers, ρ is the density of the material, N_0 is the Avogadro's constant, C_i and A_i are the weight fraction and atomic weight of element i , respectively, σ_{el}^i is the total elastic cross-section of element i , obtained from pre-calculated tabulated values^{10, 11}. Inelastic scattering is assumed to cause negligible deviation in electron trajectory, and all energy loss events are grouped into a continuous energy loss function¹². The energy loss between collisions, in keV, is calculated as

$$\begin{aligned} E_{i+1} &= E_i + \frac{dE}{dS} L, \\ \frac{dE}{dS} &= -\frac{7.85 \times 10^{-3} \rho}{E_i} \sum_{j=1}^n \frac{C_j Z_j}{F_j} \ln \left[1.116 \left(\frac{E_i}{J_j} + k_j \right) \right], \end{aligned} \quad (12)$$

where $\frac{dE}{dS}$ is the energy loss per unit length in keV/nm, Z_j is atomic number of element j , C_j is mean ionization potential of element j , k_j is a variable only dependent of Z_j ¹³. The elastic collision angle is determined using precalculated values of the partial elastic cross-section and a uniformly distributed random number¹¹.

S3. X-ray spectra measurement and data analysis

All experimental data presented in the main text were collected using the JEOL 7800 field emission scanning electron microscope (FESEM). The electron beam was directed onto the crystal surface. Crystalline thin films were mounted on gold transmission electron microscopy grids, while bulk crystals were affixed directly to a nickel scanning electron microscope holder using aluminum stubs and secured with carbon disc and carbon tapes. X-ray radiation was detected using the Oxford UltimMax 170 energy-dispersive X-ray spectroscopy (EDS) detector. The detector has a 29° take-off angle, corresponding to a center-to-center observation polar angle of $\theta_{\text{obs}} = 119^\circ$, the definition of θ_{obs} is illustrated in Supplementary Fig. 1. The distance between the EDS detector and the electron beam spot during measurements was 41.95 mm, resulting in a detector solid angle of 0.066 steradians and an observation polar angle variation of $\Delta\theta_{\text{obs}} = 16.6^\circ$.

The X-ray spectra were collected with a live time of 15 to 20 minutes and a dead time of less than 35%. The energy-dispersive X-ray spectroscopy detection range was set from 0 to 10 keV, divided into 2048 channels. The energy resolution determination of the EDS detector and spectrum calibration was performed using various characteristic X-ray peaks (details provided in Section S4). As described by Supplementary Equation (9), the tunable X-ray energy is a function of polar observation angle θ_{obs} . Consequently, both the resolution of the EDS detector and the variation in the observation polar angle influence the resulting tunable X-ray spectrum. All tunable X-ray spectra presented in this work incorporate these factors. The final spectrum was obtained by convolving the emission spectrum with the detector response, modeled as a Gaussian function with unit area and a specified full width at half maximum ($\text{FWHM}_{\text{total}}$)

$$\text{FWHM}_{\text{total}} = \sqrt{\text{FWHM}_{\Delta\theta_{\text{obs}}}^2 + \text{FWHM}_{\text{EDS}}^2}, \quad (13)$$

where FWHM_{EDS} represents the detector resolution measured by using characteristic peaks, $\text{FWHM}_{\Delta\theta_{\text{obs}}}$ represents the observation polar angle variation $\Delta\theta_{\text{obs}}$ given by

$$\begin{aligned} \text{FWHM}_{\Delta\theta_{\text{obs}}} &= \frac{2\sqrt{2\ln(2)}}{3} \frac{\partial E_p}{\partial \theta_{\text{obs}}} \Delta\theta_{\text{obs}}, \\ \frac{\partial E_p}{\partial \theta_{\text{obs}}} &= -E_p \frac{-\cos \phi_{\text{obs}} \cos \theta_{\text{obs}} v_x - \sin \phi_{\text{obs}} \cos \theta_{\text{obs}} v_y + \sin \theta_{\text{obs}} v_z}{c - \cos \phi_{\text{obs}} \sin \theta_{\text{obs}} v_x - \sin \phi_{\text{obs}} \sin \theta_{\text{obs}} v_y - \cos \theta_{\text{obs}} v_z}, \end{aligned} \quad (14)$$

where θ_{obs} and ϕ_{obs} are polar and azimuthal angle of the observation vector $\hat{\mathbf{n}}$, v_x , v_y and v_z are x , y , z components of electrons' velocity. The spectrum broadening effect caused by electron scattering was accounted for by combining the X-ray intensity formula Supplementary Equation (2) with Monte Carlo simulations, as described in Sections S1 and S2. The electron beam divergence of the FESEM (~ 1 mrad) was deemed negligible. The beam current was determined by the FESEM probe current, and the current fluctuation during the measurement time ($< 1\%$) was also negligible.

The bremsstrahlung background in the experimental data was subtracted using the NIST DTSA II software^{14–18} through a numerical method¹⁹. For all X-ray spectra presented in this study, the vertical error bars represent the standard deviation calculated from multiple measurements, while the horizontal error bars reflect the uncertainty associated with the EDS channel resolution, which is 5 eV/channel.

S4. Energy-dispersive X-ray spectroscopy (EDS) resolution and calibration

All experimental data presented in the main text and Supplementary Information were collected using the Oxford UltimMax 170 energy-dispersive X-ray spectroscopy (EDS) detector. To ensure measurement accuracy, the EDS spectrum was calibrated by detecting the characteristic peaks of C, O, F, Na, Mg, Al, and Si and aligning the peak X-ray energy with their corresponding theoretical values. The spectrum adjustment was performed using the calibration function

$$E_{\text{calib}} = 6.26 \times 10^{-6} E_{\text{raw}}^2 + 0.981 E_{\text{raw}} + 12.4, \quad (15)$$

where E_{calib} is the energy after calibration and E_{raw} is the measured energy from EDS detector. Prior to calibration, the observed values differ from their theoretical values up to 7.4 eV, and after calibration all values match their corresponding theoretical values within an accuracy of 1 eV, as shown in the Supplementary Table 1.

Supplementary Table 1. EDS detector calibration in the soft X-ray energy regime.

Element	E_{theory} [eV]	E_{raw} [eV]	$ E_{\text{theory}} - E_{\text{raw}} $ [eV]	E_{calib} [eV]	$ E_{\text{theory}} - E_{\text{calib}} $ [eV]
C	277.0	269.5	7.4	277.3	0.3
O	524.9	520.1	4.9	524.4	0.5
F	676.8	674.0	2.8	676.7	0.1
Na	1041.0	1041.5	0.5	1041.2	0.2
Mg	1253.6	1254.9	1.3	1253.7	0.1
Al	1486.7	1488.3	1.6	1486.8	0.1
Si	1740.0	1741.0	1.0	1739.8	0.2

The energy resolution FWHM_{EDS} of the EDS detector varies with the X-ray energy. In this study, the FWHM values were measured from characteristic peaks of C, Fe, Ni, Na, Al, and Si. These values were then used to fit the equation provided in the literature²⁰ to determine the energy resolution of the Oxford UltimMax 170 detector in the soft X-ray regime. The intrinsic linewidths of the characteristic peaks were accounted for and modeled using Lorentzian functions²¹. The energy resolution FWHM_{EDS} as a function of X-ray energy is given by

$$\text{FWHM}_{\text{EDS}} = \sqrt{2.52 E_p + 988}, \quad (16)$$

where E_p is the X-ray energy and the number 2.52 and 988 are fitting parameters acquired based on Supplementary Table 2.

Supplementary Table 2. EDS energy resolution in the soft X-ray energy regime.

Element	E_p [eV]	Intrinsic FWHM [eV]	Measured FWHM [eV]	Fitting results [eV]
C	277.0	7.80	40.4	41.0
Fe	705.0	2.80	52.6	52.6
Ni	851.5	2.90	57.3	55.9
Na	1041.0	0.30	60.4	60.1
Mg	1253.6	0.36	64.5	64.4
Al	1490.1	0.42	68.6	68.8
Si	1743.1	0.48	72.5	73.3

S5. X-ray self-absorption of different van der Waals materials

When X-ray emission occurs from an electron trajectory within the crystal, the emitted photons are partially absorbed by the crystal itself. The degree of self-absorption depends on the absorption path length and the material's property. It is assumed that the X-ray radiation intensity is uniformly emitted along a single electron trajectory between two collision events. The average X-ray absorption of a single electron trajectory is calculated using the Beer-Lambert law

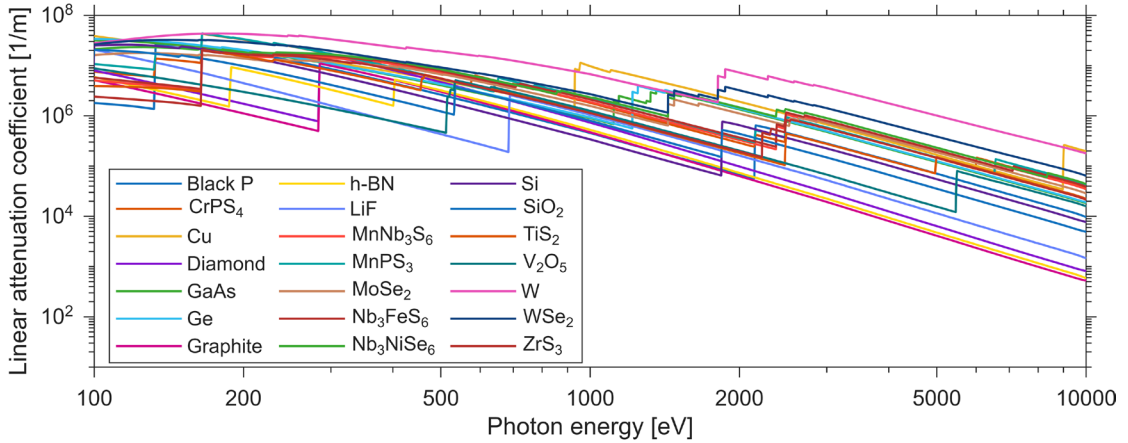
$$I = I_0 e^{-\alpha L}, \quad (17)$$

where I_0 is the radiation intensity of a single the electron trajectory without absorption, I is the radiation intensity after absorption, α is the X-ray linear attenuation coefficient which depends on the material property and X-ray photon energy, and L is the X-ray absorption distance of one electron trajectory which depends on the electron position and the crystal geometry and orientation. In our regime of study, the electron position

variation between two consecutive collisions is small compared with the absorption path length. The value of L is calculated based on the middle position of the electron trajectory using the following equation

$$L = \frac{[(x_{\text{mid}}, y_{\text{mid}}, z_{\text{mid}}) - (x_{\text{surf}}, y_{\text{surf}}, z_{\text{surf}})] \cdot \widehat{\mathbf{n}}_{\text{surf}}}{\widehat{\mathbf{n}} \cdot \widehat{\mathbf{n}}_{\text{surf}}}, \quad (18)$$

where $(x_{\text{mid}}, y_{\text{mid}}, z_{\text{mid}})$ is the middle position of electron trajectory in the coordinate system, $(x_{\text{surf}}, y_{\text{surf}}, z_{\text{surf}})$ is any point on the crystal surface in the coordinate system, $\widehat{\mathbf{n}}_{\text{surf}}$ is the unit of vector of the normal vector of the material surface plane, and $\widehat{\mathbf{n}}$ is the observation direction. The X-ray linear attenuation coefficient α for different materials are obtained from the National Institute of Standards and Technology (NIST) database²². The α value of all crystals we considered in our work with photon energy from 100 eV – 10 keV are shown in the Supplementary Fig. 2.

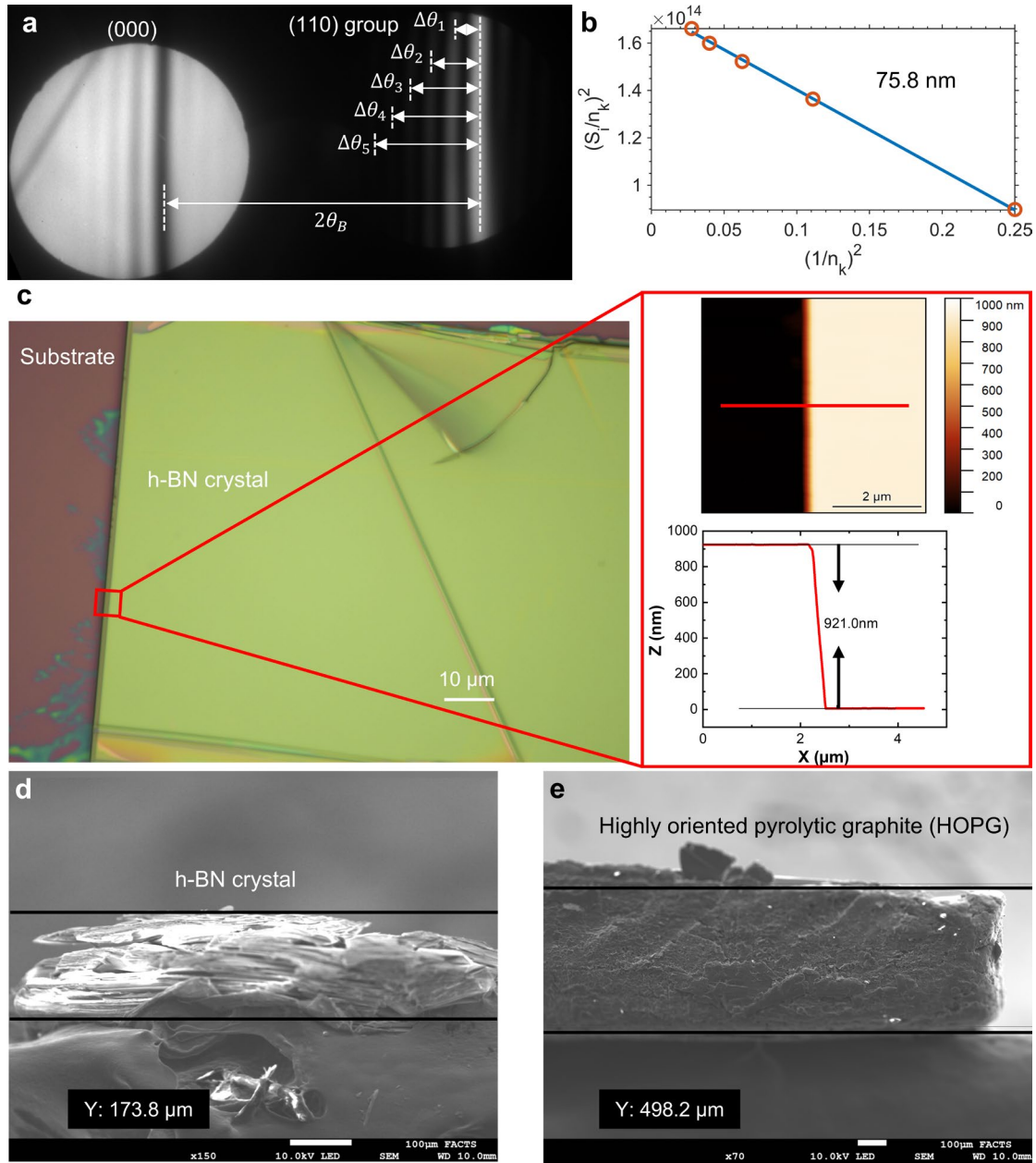


Supplementary Fig. 2. The X-ray linear attenuation coefficient value of different crystals from 100 eV to 10 keV photon energy.

S6. Crystal thickness measurement and orientation determination

The crystal thicknesses were measured using different techniques depending on the thickness range. For crystals thicker than 1 μm , the thickness was directly determined from cross-sectional images obtained via scanning electron microscopy (SEM). For crystals with thicknesses between 300 nm and 1 μm , atomic force microscopy (AFM) was used. For thinner crystals (< 300 nm), thickness measurements were conducted

using convergent beam electron diffraction (CBED) patterns^{23, 24} via transmission electron microscopy (TEM) JEOL 2010 HR.



Supplementary Fig. 3. **a** shows the convergent beam electron diffraction (CBED) pattern of graphite crystal acquired from TEM under 200 keV electron energy. One plane of (110) plane group is strongly excited under the two-beam condition. **b** shows the thickness determination by using Supplementary Equation (19) and (20). **c** h-BN crystal thickness measurement using atomic force microscopy (AFM). **d** h-BN crystal thickness measured from cross-sectional images obtained via scanning electron microscopy (SEM). **e** The same scenario as **d** but for HOPG crystal.

Detailed explanation of the relationship between Kossel-Möllenstedt (K-M) fringes and crystal thickness can be found in the literature²⁵. The Deviation S_i for the i th fringe is given by:

$$S_i = \lambda \frac{\Delta\theta_i}{2\theta_B d^2}, \quad (19)$$

where λ is the electron wavelength, θ_B is the Bragg angle for the diffracting hkl plane, d is the hkl interplanar spacing. The crystal thickness t can be determined by

$$\frac{S_i^2}{n_k^2} + \frac{1}{\xi_g^2 n_k^2} = \frac{1}{t^2}, \quad (20)$$

where n_k is an integer series 1,2,3... and ξ_g is the extinction distance. Without knowing the extinction distance ξ_g , we vary n_k value for S_1 from $n_k = 1$ and increase until the plot of $\frac{1}{n_k^2}$ vs. $\frac{S_i^2}{n_k^2}$ is a line. An example of using this method to measure a graphite crystal with the thickness of 75.8 nm is shown in Figs. S3a and S3b.

The crystal thin film thickness value presented in this work is acquired by multiple measurements around the region of interest and taking the average. Detailed data are shown in the Supplementary Table 3a and Table 3b.

Supplementary Table 3a. The thickness measurement of graphite and h-BN crystals.

Graphite thin film 1	Graphite thin film 2	Graphite thin film 3	h-BN thin film 1	h-BN thin film 2	h-BN thin film 3
24.4 nm	69.8 nm	134.9 nm	37.9 nm	99.5 nm	208.9 nm
25.7 nm	72.2 nm	138.3 nm	38.8 nm	110.4 nm	210.3 nm
29.0 nm	75.8 nm	154.1 nm	43.4 nm	110.7 nm	213.2 nm
29.2 nm	81.2 nm	158.1 nm	43.6 nm	110.9 nm	213.3 nm
39.1 nm	81.7 nm	164.1 nm	44.6 nm	111.1 nm	248.6 nm
Ave. 29 nm StD. 6 nm	Ave. 76 nm StD. 5 nm	Ave. 150 nm StD. 13 nm	Ave. 42 nm StD. 3 nm	Ave. 109 nm StD. 5 nm	Ave. 219 nm StD. 17 nm

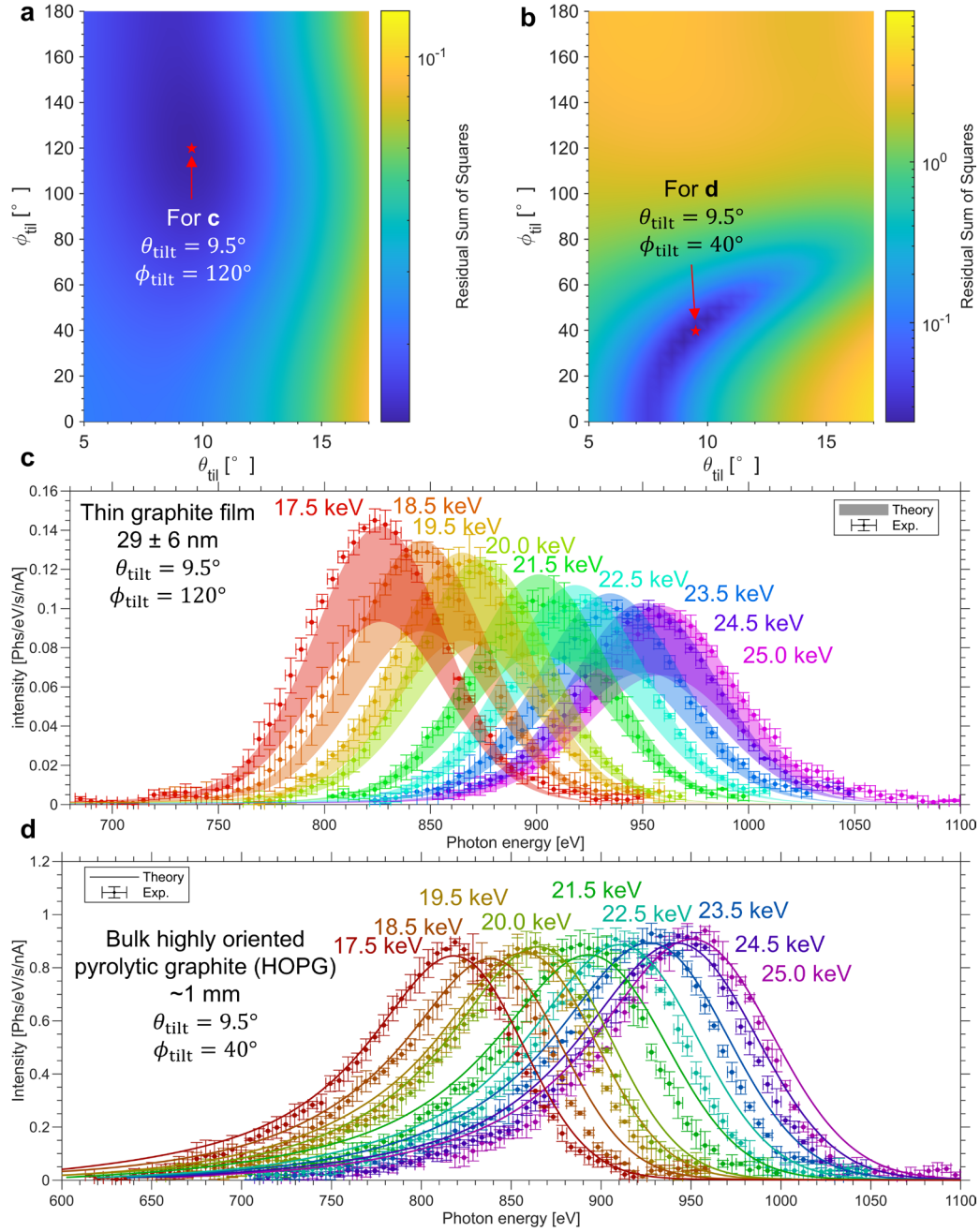
Supplementary Table 3b. The thickness measurement of MoSe₂ and WSe₂ crystals.

MoSe ₂ thin film 1	MoSe ₂ thin film 2	MoSe ₂ thin film 3	WSe ₂ thin film 1	WSe ₂ thin film 2	WSe ₂ thin film 3
43.3 nm	87.7 nm	112.0 nm	31.9 nm	50.5 nm	66.6 nm
44.3 nm	96.2 nm	141.7 nm	43.2 nm	50.5 nm	72.1 nm
48.2 nm	97.3 nm	156.7 nm	44.5 nm	52.5 nm	72.8 nm
48.3 nm	129.2 nm	161.3 nm	44.7 nm	59.1 nm	74.3 nm
51.8 nm	148.8 nm	163.2 nm	44.9 nm	60.0 nm	86.9 nm
Ave. 47 nm StD. 3 nm	Ave. 112 nm StD. 16 nm	Ave. 147 nm StD. 21 nm	Ave. 42 nm StD. 6 nm	Ave. 55 nm StD. 5 nm	Ave. 75 nm StD. 7 nm

During the X-ray measurements from SEM, crystal thin films were mounted on gold transmission electron microscopy grids, while bulk crystals were affixed directly to a nickel scanning electron microscope holder using aluminum stubs and secured with a carbon disc and carbon tapes. Due to experimental conditions such as crystal surface roughness, uneven SEM holder, and sample bending, the crystal tilt angles θ_{til} and ϕ_{til} are normally not zero and vary during each experimental session. The exact tilt angles θ_{til} and ϕ_{til} for different measurements are determined by calculating the residual sum of squares (RSS) between the experimental data and theoretical predictions. To ensure the accuracy of this method, multiple tunable X-ray spectra are collected at 9 different electron kinetic energies during one experimental session. For each electron kinetic energy, 3 spectra are collected to calculate the mean intensity of the spectrum and the y-axis error bars. The crystal tilt angles θ_{til} and ϕ_{til} are chosen to minimize the residual sum of squares (RSS) between the experimental data and theoretical predictions, and this value provides a good match for all 27 spectra. The RSS value and spectra of the 29 nm graphite thin film and bulk highly oriented pyrolytic graphite (HOPG) are shown in Supplementary Fig. 4. The values of θ_{til} and ϕ_{til} for all spectra presented in this work are shown in Supplementary Table 4.

Supplementary Table 4. The initial tilt angle θ_{til} and ϕ_{til} of different samples.

Sample	Polar angle θ_{til}	Azimuthal angle ϕ_{til}
Graphite film 29 ± 6 nm	9.5°	120°
Graphite film 76 ± 5 nm	13.0°	120°
Graphite film 150 ± 13 nm	6.5°	180°
Bulk graphite (HOPG) ~ 17 μm	11.0°	60°
Bulk graphite (HOPG) ~ 500 μm	11.5°	40°
Bulk graphite (HOPG) ~ 1 mm	9.5°	40°
h-BN film 42 ± 3 nm	13.5°	115°
h-BN film 219 ± 17 nm	11.5°	65°
h-BN film 109 ± 5 nm	13.5°	130°
h-BN film 219 ± 17 nm	17.0°	130°
h-BN film ~ 659 nm	14.5°	105°
h-BN film ~ 921 nm	17.0°	130°
Bulk h-BN ~ 170 μm	20.0°	65°



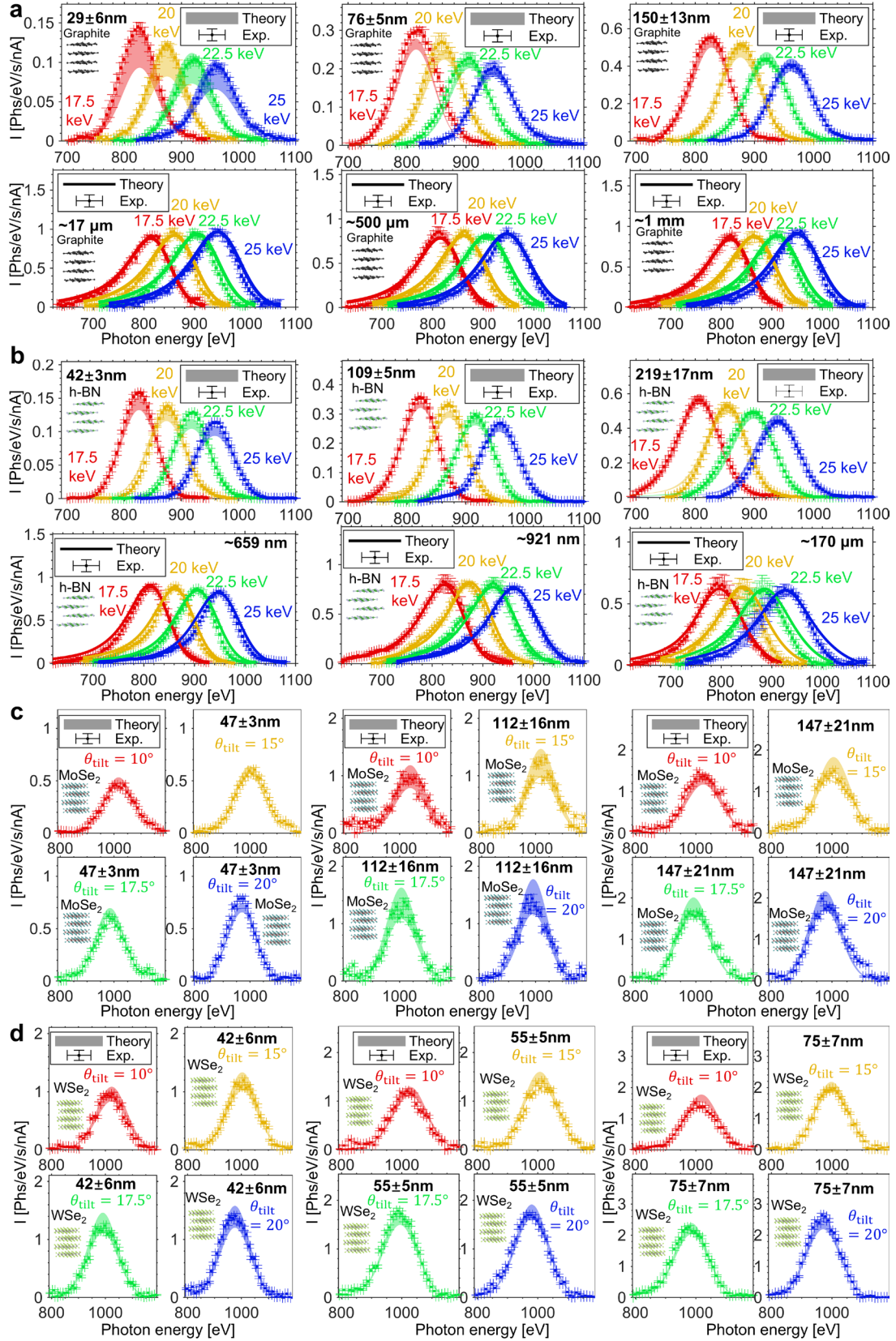
Supplementary Fig. 4. **a** Residual sum of squares (RSS) for a graphite thin film with a thickness of 29 nm, acquired by comparing experimental data and theoretical predictions at various values of θ_{til} and ϕ_{til} . The red pentagram marks the location of the minimum RSS. **b** RSS for highly oriented pyrolytic graphite (HOPG) bulk crystal under the same conditions as in **a**. **c** Comparison of experimental data and theoretical predictions at the red pentagram point identified in **a**. Translucent bands indicate variations in intensity due to uncertainties in the thin film thickness under experimental conditions. **d** Comparison of experimental data and theoretical predictions at the red pentagram point identified in **b**. In panels **c** and **d**, the y-axis unit is photons/eV/s/nA, and all spectra correspond to a solid angle of 0.066 steradians, which is in accordance with experimental conditions.

S7. Tunable X-ray spectra from different crystals, thicknesses and setups

Supplementary Fig. 5 presents the tunable X-ray spectra for all van der Waals crystals discussed in this work, measured across a wide range of thicknesses, electron energies, crystal tilt angles, and electron sources. The experimental results are shown alongside theoretical predictions, demonstrating a good agreement between theory and experiment.

Spectra for graphite and h-BN were acquired using a field emission scanning electron microscope (FESEM) JEOL 7800 at four electron energies: 17.5, 20, 22.5, and 25 keV. While spectra for MoSe₂ and WSe₂ were obtained using a transmission electron microscope (TEM) JEOL 2010 HR at four tilt angles: 10°, 15°, 17.5° and 20° with a fixed electron energy of 200 keV. The X-ray spectra were acquired using an energy-dispersive X-ray spectroscopy (EDS) detector in both SEM and TEM experiments. For SEM measurements, spectra were collected with a live time of 15 to 20 minutes and a dead time less than 35%. For TEM measurements, spectra were collected with a live time of 30 minutes and a dead time less than 45%. The X-ray spectral data analysis, including energy calibration and EDS resolution, follows the same methodology described in Supplementary Information Section 4.

Graphite, h-BN, MoSe₂ and WSe₂ crystals were mechanically exfoliated onto silicon substrates with a 285 nm SiO₂ layer and subsequently transferred to Au TEM grids using a wet-transfer method. The thickness measurement methods for graphite, h-BN, MoSe₂ and WSe₂ crystals are described in Supplementary Information Section 6 crystal thickness measurement part, and the corresponding thickness values are listed in Supplementary Table 3a and 3b. The crystal initial tilt angle θ_{til} and polar ϕ_{til} for h-BN and graphite are described in Section 6 orientation determination part. For MoSe₂ and WSe₂, the crystal tilt angle θ_{til} is controlled by first aligning the electron beam along the [001] zone axis using Kikuchi lines, followed by tilting the crystal using a double-tilt TEM holder.

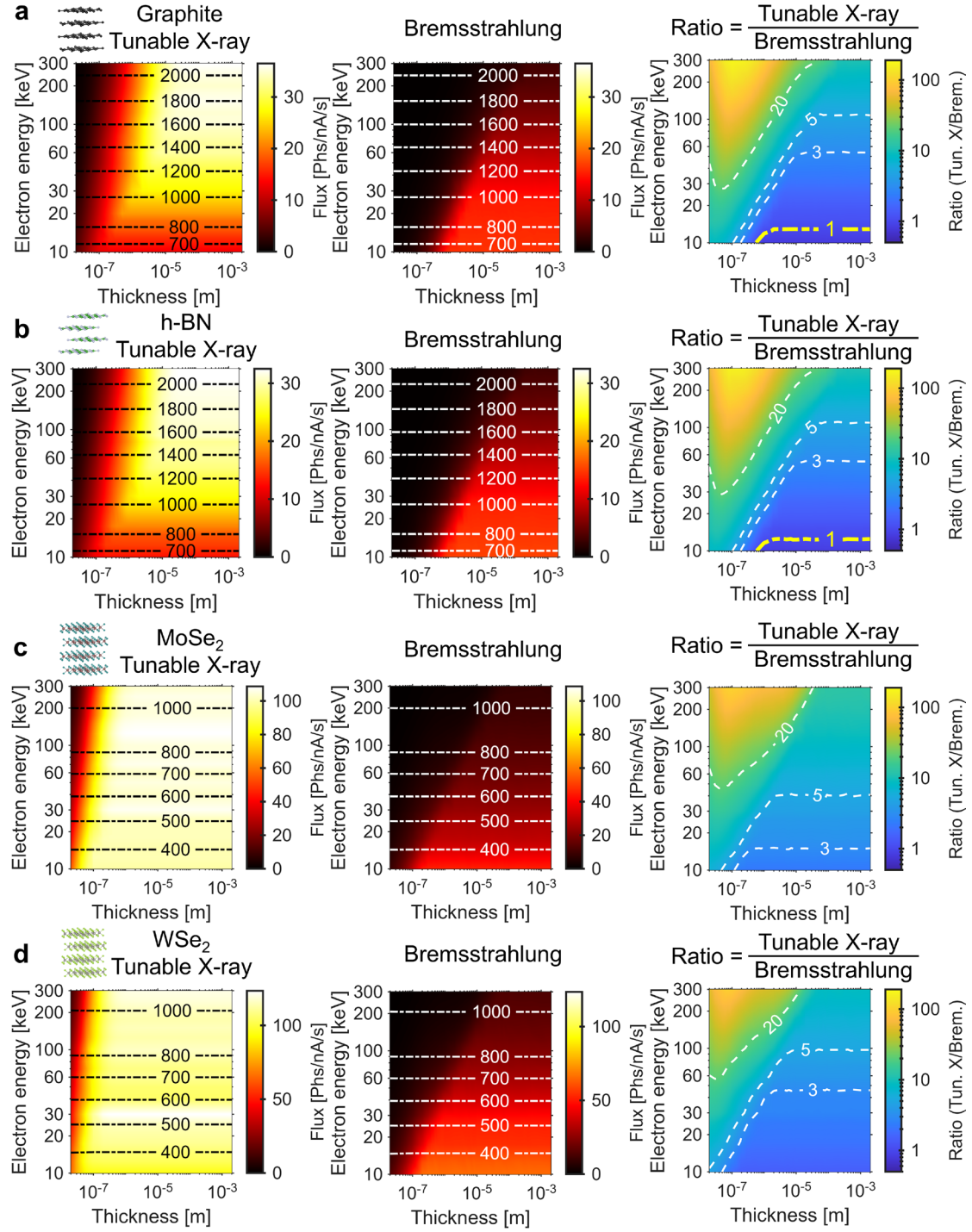


Supplementary Fig. 5. **a** Tunable X-ray emission from graphite of varying thicknesses and electron energies, measured using SEM. Translucent bands represent intensity variations due to uncertainties in thin film. **b** The same scenario as **a** but for h-BN. **c** Tunable X-ray emission from MoSe₂ with different thicknesses and crystal tilt angles. Acquired from TEM with 200 keV electron energy. **d** The same scenario as **c** but for WSe₂. All units, data processing methods and theory values are in accordance with the main text.

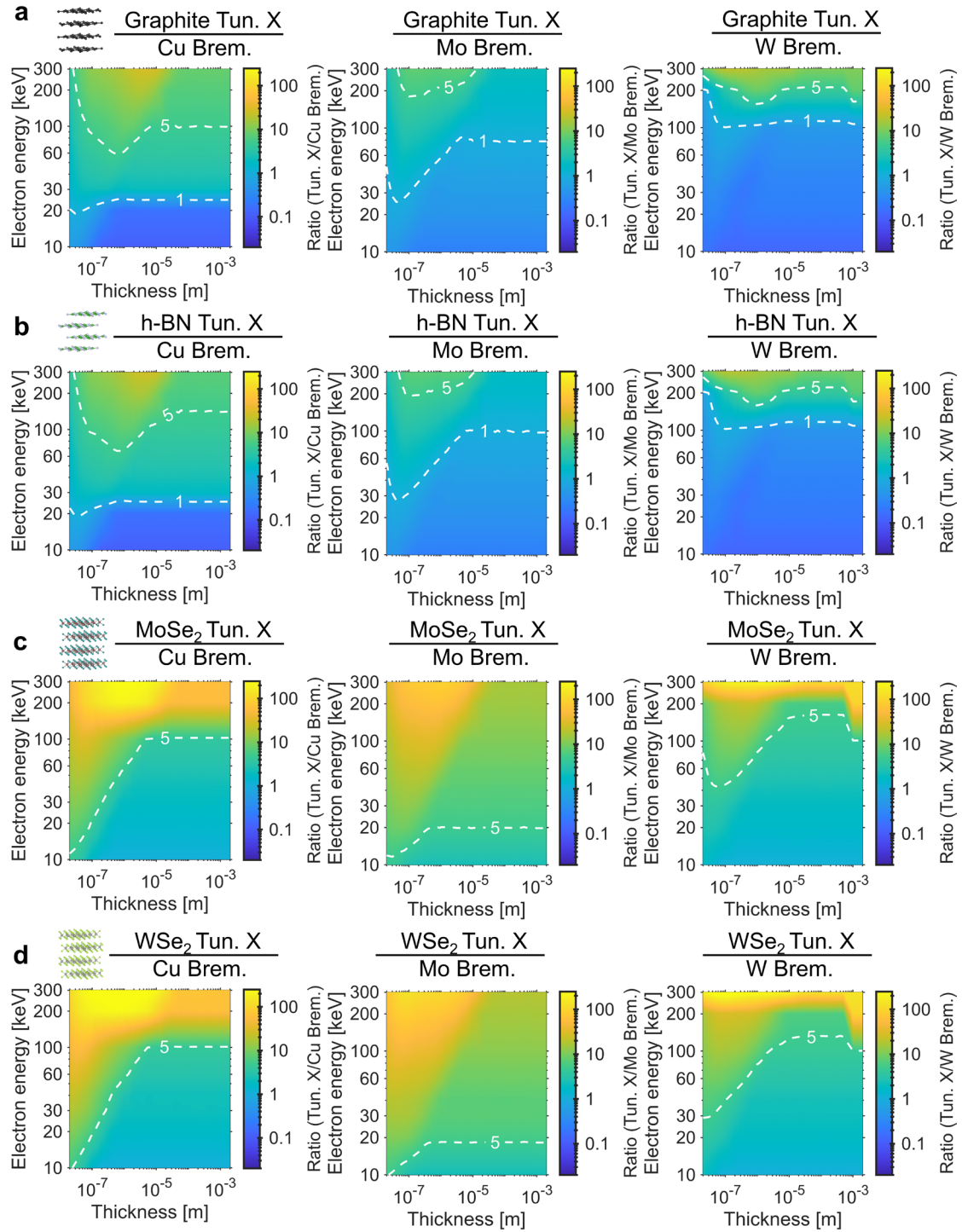
S8. Comparison of tunable X-ray flux with bremsstrahlung

The comparison between the tunable X-ray and bremsstrahlung flux for different vdW materials demonstrated in the main text only covers the bulk crystal condition with 2 mm material thickness and at 300 keV electron kinetic energy. A more comprehensive picture is presented in Supplementary Fig. 6, where the tunable X-ray flux, bremsstrahlung flux and the ratio between above two quantities are given at different crystal thicknesses ranging from nanometer to millimeter and electron kinetic energies ranging from 10 keV to 300 keV. In Figs. S6a and S6b, the critical points at which bremsstrahlung and tunable X-ray flux are equal are indicated by yellow dashed lines labeled with the number 1, representing a flux ratio of unity. These critical points typically occur at low electron energies and large crystal thicknesses. For a given electron energy, the ratio between bremsstrahlung and tunable X-ray flux initially changes with thickness but tends to stabilize beyond a certain thickness. Notably, throughout the studied regime, tunable X-ray emission from MoSe₂ and WSe₂ consistently exceeds that of bremsstrahlung.

To this day, most X-ray imaging and spectroscopy systems still employ conventional electron-impact X-ray tubes as their sources. Their emission spectrum includes high-intensity characteristic X-rays along with a broad-band but low intensity bremsstrahlung background. In Supplementary Fig. 7, we compare our tunable X-ray flux with the bremsstrahlung generated from three different materials: Cu, Mo and W which are commonly used as X-ray tube anode materials.



Supplementary Fig. 6. **a** Shows the relationship among tunable X-ray flux, bremsstrahlung flux, material thickness, and electron kinetic energy for graphite. In the left panel, the colormap provides a comprehensive view of tunable X-ray flux as a function of both material thickness and electron kinetic energy. The X-axis represents material thickness, while the Y-axis represents electron kinetic energy; color intensity indicates the tunable X-ray flux value in units of photons/s/nA. The dashed lines with numbers indicate the tunable X-ray peak photon energies. In the middle panel, the colormap shows the comprehensive view of bremsstrahlung flux as a function of both material thickness and electron kinetic energies calculated using the Monte Carlo method¹⁴⁻¹⁸. In the right panel, the ratio between the tunable X-ray and the bremsstrahlung flux at different material thickness and electron kinetic energies are shown. The dashed lines with numbers indicate the ratio value. **b** Demonstrates the same scenario of the material h-BN. **c** Demonstrates the same scenario of the material MoSe₂. **d** Demonstrates the same scenario of the material WSe₂. All flux values are calculated from non-tilted crystals by integrating the intensity spectrum within 10 eV bandwidth of the peak and correspond to a solid angle of 0.066 steradians.

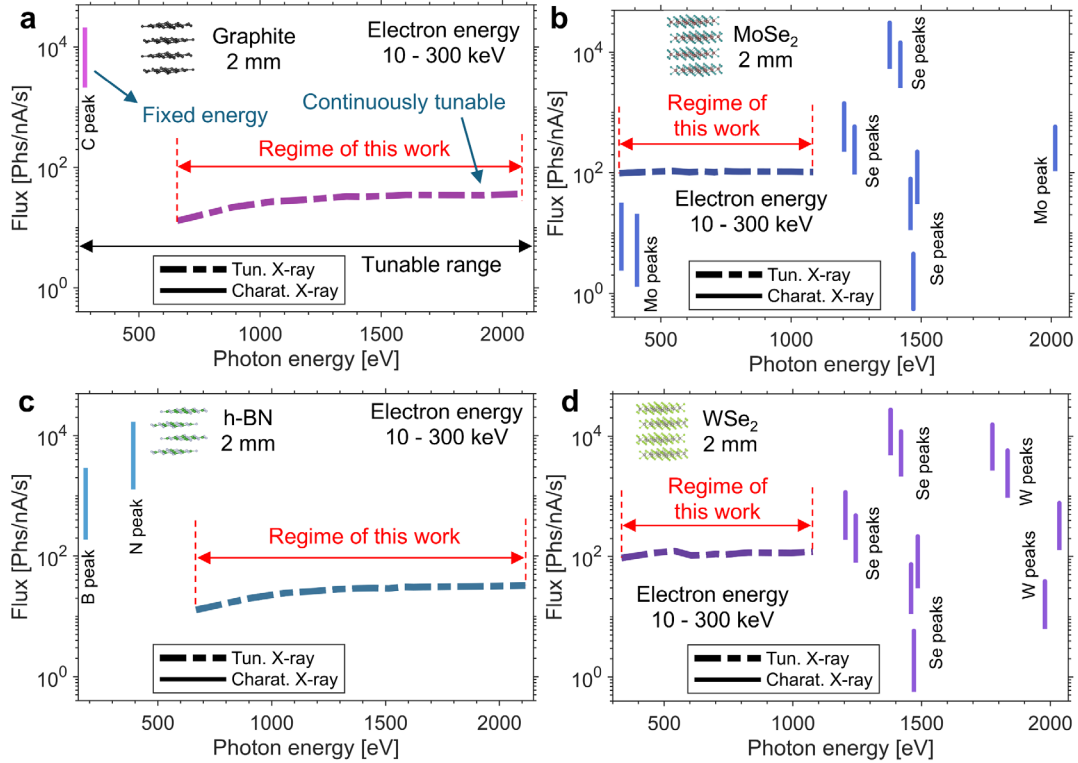


Supplementary Fig. 7. **a** Comparison of graphite tunable X-ray flux and bremsstrahlung flux as a function of material thickness and electron kinetic energy. The left panel shows the ratio between graphite tunable X-ray flux and bremsstrahlung flux from Cu, with color intensity representing the ratio values. Dashed lines with labels indicate the positions where the ratio equals 1 and 5. The middle panel presents the colormap of the ratio between graphite tunable X-ray flux and bremsstrahlung flux from Mo, while the right panel shows the same for W. **b** Demonstrates the same scenario of the material h-BN. **c** Demonstrates the same scenario of the material MoSe₂. **d** Demonstrates the same scenario of the material WSe₂. All flux values are derived from non-tilted crystals, calculated by integrating the intensity spectrum within a 10 eV bandwidth of the peak, and correspond to a solid angle of 0.066 steradians. Bremsstrahlung flux values are computed using the Monte Carlo method^{14–18}.

S9. Comparison of tunable X-ray flux with characteristic peaks

In this section, we compare the photon flux of tunable X-rays generated from four vdW crystals – graphite, h-BN, MoSe₂, and WSe₂ – with that of characteristic X-ray peaks from the same materials, over a range of electron kinetic energies from 10 keV to 300 keV. The flux ratio between tunable and characteristic X-rays varies significantly with both electron energy and the position of characteristic peak. The flux of the C K_{α} peak at 277 eV is 59 times higher than the tunable X-ray. Conversely, under the same electron energy and a 2 mm WSe₂ thickness, the tunable X-ray intensity can exceed the Se L peak intensity at 1470 eV by a factor of 211. However, unlike characteristic peaks which are fixed in energy, tunable X-rays offer real-time spectral tunability. This tunability enables efficient filling of gaps between characteristic lines, while delivering intensities far exceeding bremsstrahlung.

Supplementary Fig. 8 illustrates this capability, underscoring the versatility of tunable X-rays for diverse applications. All flux values are calculated using a Monte Carlo method^{14–18} consistent with other flux evaluations in this study.



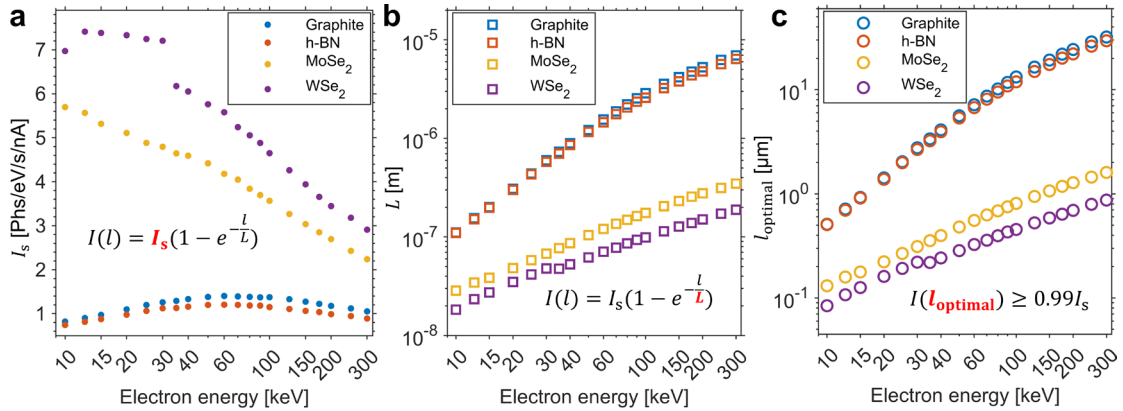
Supplementary Fig. 8. a Comparison of tunable X-ray flux and characteristic X-ray flux for 2 mm thick graphite. Solid lines represent characteristic X-ray peaks at different electron energies, where photon energy remains constant but flux varies. Dashed lines represent tunable X-rays, where both photon energy and flux vary with electron energy. b The same scenario for MoSe₂. c The same scenario for h-BN. d The same scenario for WSe₂. All flux values are derived from non-tilted crystals, calculated by integrating the intensity spectrum within a 10 eV bandwidth of the peak, and correspond to a solid angle of 0.066 steradians.

S10. Fitting parameters and optimal crystal thicknesses

The inset in Fig. 3a of the main text illustrates the relationship between tunable X-ray intensity and crystal thickness under our experimental conditions. This relationship is described by the fitting equation

$$I(l) = I_s(1 - e^{-\frac{l}{L}}), \quad (21)$$

where I is the tunable X-ray intensity, l is the crystal thickness, I_s is the saturation tunable X-ray intensity as the crystal thickness increases, and L is the fitting parameter. The values of I_s and L depend on the material properties and electron energy. Figs. S9a and S9b present the values of I_s and L for four different materials across various electron energies. In addition, Supplementary Fig. 9c shows the optimal crystal thickness l_{optimal} , defined as the thickness at which 99% of the saturation intensity I_s is reached, for each material and energy condition. The simulation is performed by using our theoretical framework introduced in the main text, which accounts for electron scattering and X-ray self-absorption and verified with experimental results.



Supplementary Fig. 9. **a** Saturation tunable X-ray intensity I_s for four materials: Graphite, h-BN, MoSe₂ and WSe₂, at electron kinetic energies ranging from 10 keV to 300 keV. **b** Fitting parameter L for the same materials over the same energy range. **c** Optimal material thickness l_{optimal} , defined as the thickness required to achieve 99% of the saturation intensity I_s . All values are calculated using the theoretical framework for non-tilted crystals, accounting for the EDS detector's spectral broadening, and correspond to a solid angle of 0.066 steradians.

S11. Thermal analysis

The maximum current density that can be applied to the graphite sample imposes a practical limitation on the tunable X-ray source. To assess the thermal robustness of the proposed scheme under realistic electron beam parameters, we performed a thermal analysis. The simulation was conducted on a square bulk graphite crystal with a side length of 5 mm and a thickness of 2 mm, dimensions typical for highly oriented pyrolytic graphite (HOPG). The crystal is mounted on a cylindrical copper holder with a diameter of 10 mm and a depth of 3 mm. For boundary conditions, the outer surface of the copper holder is assumed to be in contact with surrounding device components maintained at a constant temperature of 293.15 K (room temperature).

Assuming all electron energy loss within the crystal contributes to heating, the total heat power transferred from the continuous electron beam to the graphite crystal $Q(\mathbf{r})$ is calculated using a Monte Carlo simulation as follows

$$Q(\mathbf{r}) = \frac{IV}{NE_e} \sum_{j=1}^N \frac{\Delta E_j(\mathbf{r})}{\Delta V}, \quad (22)$$

Where $\mathbf{r}$ is the position vector, I is the beam current, V is the electron acceleration voltage, N is the total number of simulated electrons (100,000 in our case). E_e is the energy of a single electron, which corresponds to the acceleration voltage V . The crystal is discretized into small volumes $\Delta V = \Delta x \Delta y \Delta z$ and $\Delta E_j(\mathbf{r})$ denotes the energy lost by the j^{th} electron within the volume element at position $\mathbf{r}$. For both the Monte Carlo and thermal simulations, the spatial distribution of the electron beam $I(x, y)$ is modeled as a 2D Gaussian function in the XY-plane

$$I(x, y) = \frac{1}{2\pi\sigma^2} e^{-\left[\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}\right]}, \quad (23)$$

where σ is the standard deviation of the distribution, determined by the electron beam spot size. In our case, a beam diameter of 1 mm is used. The beam center is located at $x_0 = y_0 = 0$. Figs. S10a and S10b show the spatial distribution of the electron beam heat transfer for 300 keV electrons.

We consider two mechanisms for heat dissipation: thermal conduction and radiation. Thermal conduction is modeled by solving the anisotropic heat equation:

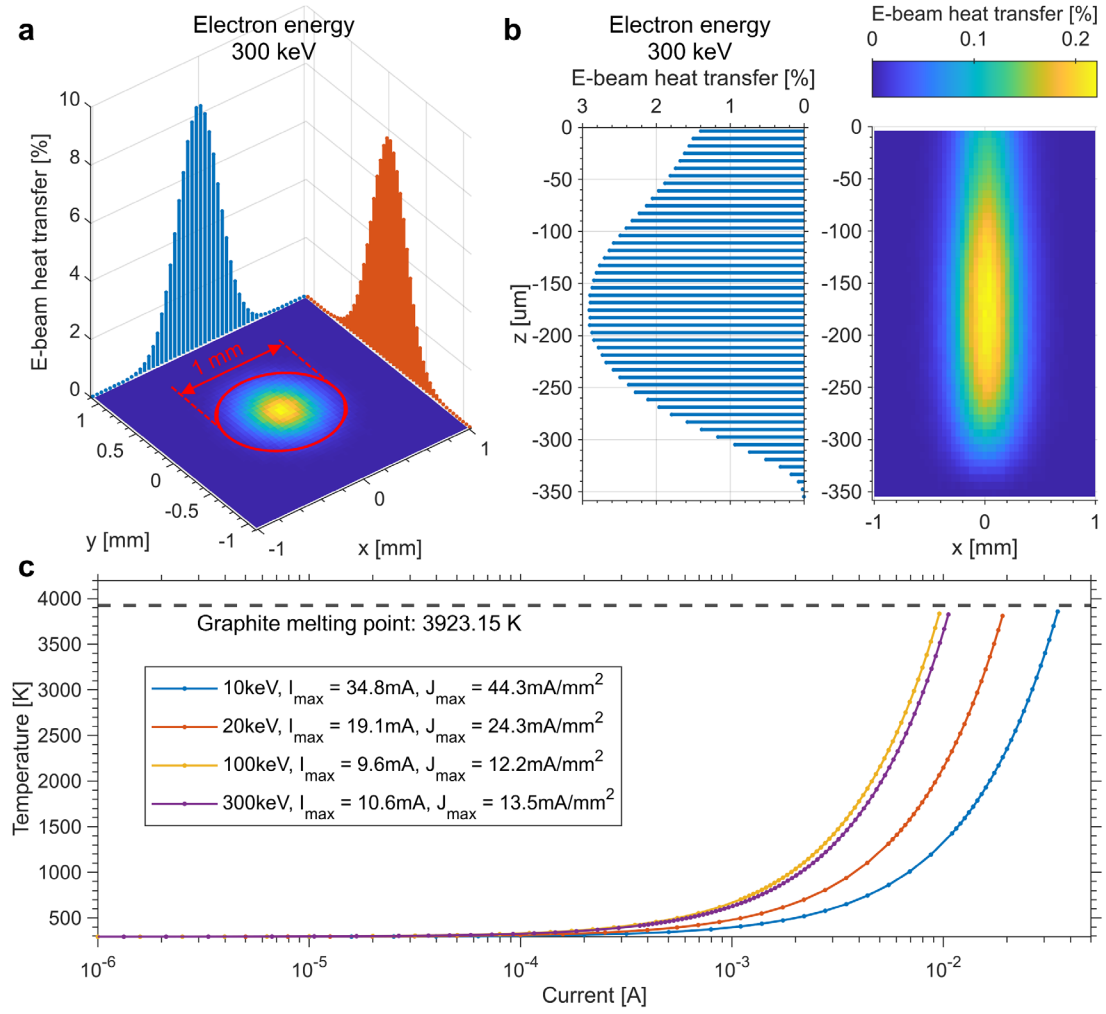
$$\rho C_p \frac{\partial T(x, y, z, t)}{\partial t} - k_{\parallel} \left[\frac{\partial^2 T(\mathbf{r}, t)}{\partial x^2} + \frac{\partial^2 T(\mathbf{r}, t)}{\partial y^2} \right] - k_{\perp} \frac{\partial^2 T(\mathbf{r}, t)}{\partial z^2} = Q(\mathbf{r}), \quad (24)$$

where ρ is the density, C_p is the specific heat capacity at constant stress, $T(\mathbf{r}, t)$ is the temperature at position $\mathbf{r}$ and time t . $k_{\parallel}$ and $k_{\perp}$ are the in-plane and out-of-plane thermal conductivities, respectively, with $k_{\parallel} \gg k_{\perp}$ for van der Waals materials such as graphite. $Q(\mathbf{r})$ is the position-dependent heat source from the incident electron beam. Thermal radiation heat loss at the surface is modeled by the Stefan–Boltzmann law

$$-\hat{\mathbf{n}} \cdot \mathbf{q} = \epsilon \sigma_s (T_0^4 - T^4), \quad (25)$$

where $\hat{\mathbf{n}}$ is the unit normal vector to the boundary surface, $\mathbf{q}$ is the heat flux vector, ϵ the material emissivity, σ_s is the Stefan-Boltzmann constant and T_0 is the ambient temperature, set to 293.15 K. The thermal simulation is conducted via COMSOL Multiphysics® software using heat transfer in solids module.

In our simulations, the maximum temperature in the graphite crystal reaches steady state within 5 seconds. Supplementary Fig. 10c presents the steady-state maximum temperature as a function of electron beam energy and current. The legend indicates the corresponding maximum current $I_{\max}$ and maximum current density $J_{\max}$ for each case. The required X-ray flux for X-ray absorption near edge structure (XANES) and X-ray emission spectroscopy (XES) ranges from 0.2 to 8×10^4 photons/s/sr/eV²⁶, while that for X-ray imaging is $0.2 - 0.4 \times 10^4$ photons/s/mm²²⁷. Our tunable X-ray source meets these requirements at beam currents of 0.4 μ A and 1.4 μ A, respectively – both significantly lower than the maximum sustainable current of 10.6 mA.

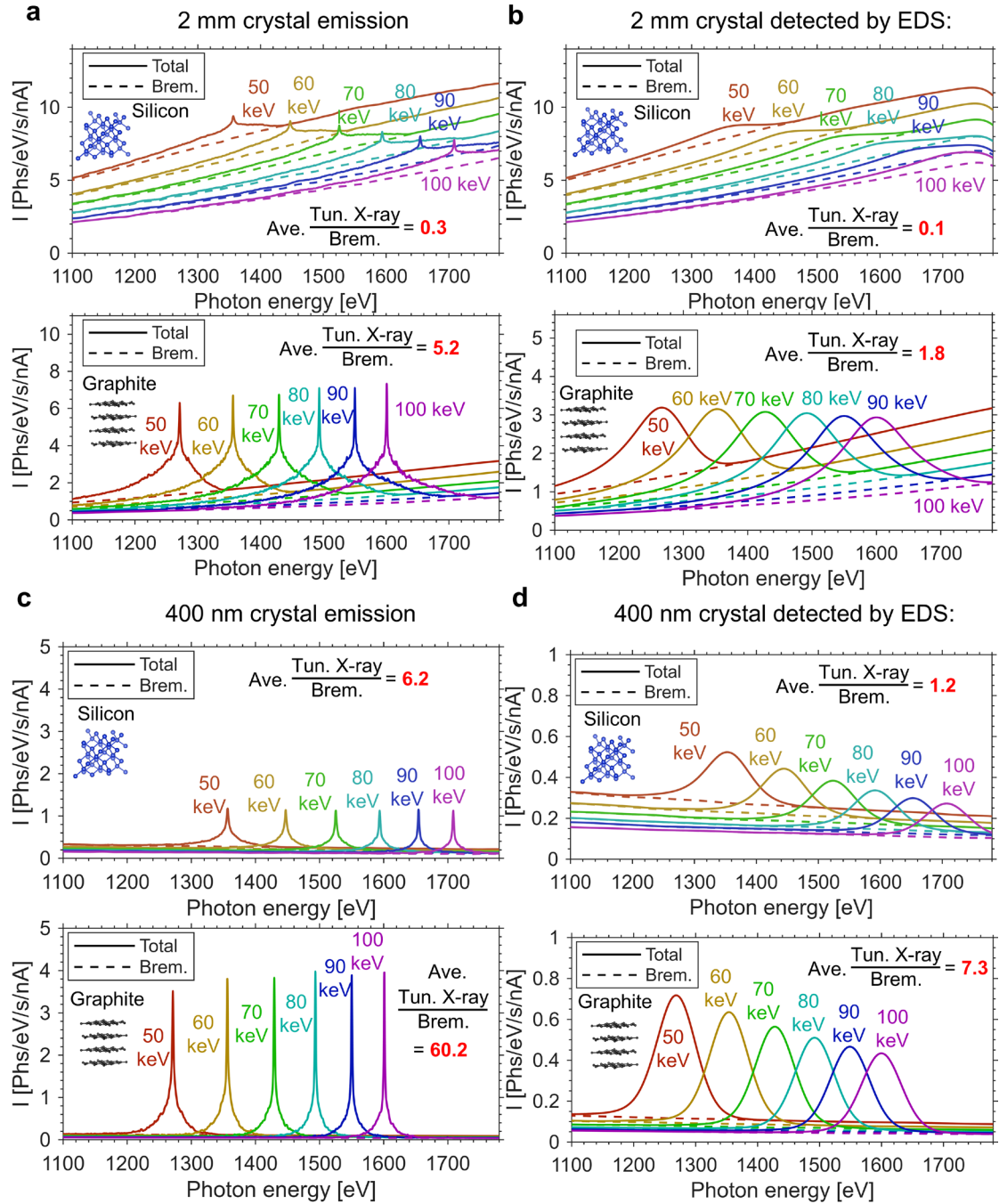


Supplementary Fig. 10. **a** Heat transfer profile on the XY-plane under a 300 keV electron beam. **b** Heat transfer profile along the Z direction and on the XZ-plane under the same beam conditions. **c** Maximum steady-state temperature of graphite as a function of electron energy and beam current. The corresponding maximum current and current density values are indicated in the legend.

S12. Comparison of tunable X-ray generation of silicon and graphite

Detecting tunable X-rays from bulk materials using tabletop electron sources has been considered experimentally challenging due to relatively high electron scattering²⁸. In this section, we provide a theoretical explanation for these earlier conclusions using our own framework. Specifically, we compare the tunable X-ray intensity with bremsstrahlung for two materials: graphite, used in our study, and silicon, used in the prior study that failed to detect tunable X-rays from bulk samples. The electron energy range is set between 50 and 100 keV, consistent with that earlier investigation. For both materials, we consider crystal planes that yield the strongest tunable X-ray emission, the (002) plane for graphite and the (111) plane for silicon.

The ratio of tunable X-ray to bremsstrahlung intensity is the key determining factor for whether tunable X-rays are experimentally observable. We consider two scenarios. In the first, a bulk crystal with a thickness of 2 mm is considered, under this condition, tunable X-rays from silicon were not observed in the previous study. The simulation results show that the tunable X-ray to bremsstrahlung ratio for silicon is only 0.3, and it further decreases to 0.1 when the energy broadening effect of the EDS detector (discussed in Supplementary Information Section S4) is considered. In the second scenario, a 400 nm-thick film is considered, a condition under which tunable X-ray emission has been observed for both materials. With EDS broadening included, the tunable X-ray to bremsstrahlung ratio is 1.2 for silicon and 7.3 for graphite, significantly higher than in the bulk case. These results explain why tunable X-ray signals are more readily detected in thin films, owing to the higher tunable-to-bremsstrahlung intensity ratio. Detailed simulation results are presented in Supplementary Fig. 11.



Supplementary Fig. 11. **a** X-ray emission spectra from 2 mm thick bulk silicon (upper panel) and graphite (lower panel) under electron energies ranging from 50 to 100 keV. Solid lines represent the total X-ray intensity, including both tunable X-rays and bremsstrahlung, while dashed lines show bremsstrahlung alone. The average ratio of tunable X-ray to bremsstrahlung intensity across all electron energies is highlighted in red. **b** The same scenario as in **a** with the spectral broadening effect of the EDS detector taken into account. **c** X-ray emission spectra from a 400 nm thin films silicon (upper panel) and graphite (lower panel) under different electron energies from 50 – 100 keV. The average value of tunable X-ray to bremsstrahlung ratio across all electron energies is highlighted in red. **d** The same scenario as in **c** with the spectral broadening effect of the EDS detector taken into account. All values are calculated using the established theoretical framework, and correspond to a solid angle of 0.066 steradians.

S13. Non-monotonic change in bremsstrahlung flux

In the main text Fig. 4b and 4c, we compare the tunable X-ray flux with bremsstrahlung flux across a wide range of crystal thicknesses. Notably, the bremsstrahlung flux exhibits a non-monotonic dependence on thickness (the bending region). This behavior arises from the combined influence of two effects: the electron scattering and X-ray self-absorption, both of which are incorporated in our bremsstrahlung simulations using a Monte Carlo method¹⁷.

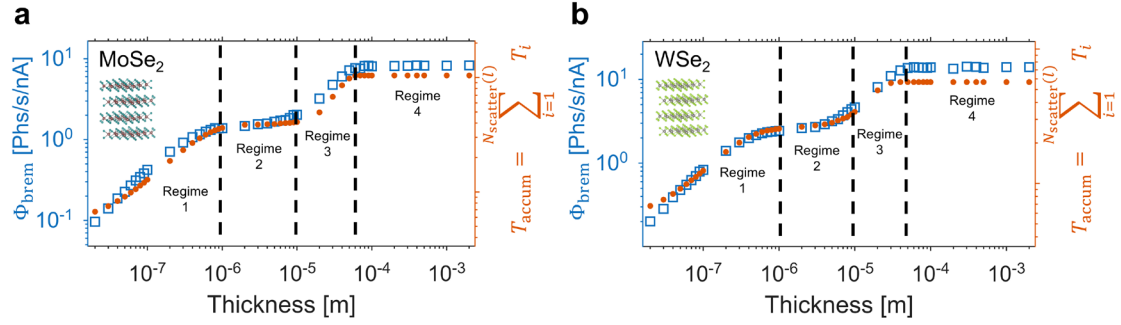
The bremsstrahlung intensity as a function of crystal thickness can be divided into four regimes: In regime 1, increasing crystal thickness leads to a longer electron-crystal interaction length, resulting in a rise in bremsstrahlung flux. In regime 2, further increases in thickness extend the interaction length, but newly generated bremsstrahlung photons come deeper within the crystal. These photons experience stronger X-ray self-absorption, limiting their ability to escape and thus saturating the flux. In regime 3, as electron scattering becomes more severe, a greater number of electrons are backscattered toward the near-surface region. These backscattered electrons still retain sufficient energy (~ 10 keV in our case) to generate bremsstrahlung photons (~ 1 keV in our case). Since this occurs closer to the surface, self-absorption is reduced, allowing more photons to escape and causing a secondary rise in flux. In regime 4, once the crystal thickness exceeds the electron interaction volume, additional increases no longer affect electron-crystal interactions. Consequently, the bremsstrahlung flux saturation.

This behavior can be quantitatively described by the accumulated transmission

$$T_{\text{accum}} = \sum_{i=1}^{N_{\text{scatter}}(l)} T_i = \sum_{i=1}^{N_{\text{scatter}}(l)} e^{-\alpha L(x_i, y_i, z_i)}, \quad (26)$$

where $N_{\text{scatter}}(l)$ is the total number of electron scattering segments (which is a function of crystal thickness l), and T_i is the transmission of the X-ray emitted from i^{th} electron scattering segment. The value is calculated by using Beer–Lambert law which is the same as Supplementary Equation (17), α is the X-ray linear attenuation coefficient which depends on the material property and X-ray photon energy, $L(x_i, y_i, z_i)$ is the X-ray absorption distance of i^{th} electron scattering segment, which

depends on the electron position (x_i, y_i, z_i) and the crystal geometry. In regime 1, the accumulated transmission increases with crystal thickness due to the rising of $N_{\text{scatter}}(l)$. In regime 2, although N_{scatter} continues to grow, the associated T_i values decrease due to greater X-ray self-absorption, leading to a plateau in the accumulated transmission. In regime 3, backscattered electron segments near the crystal surface begin to contribute, where self-absorption is minimal. As a result, the accumulated transmission increases again. In regime 4, the electron scattering profile no longer changes with additional thickness, and the accumulated transmission remains constant. Supplementary Fig. 12 demonstrates how the interplay between electron scattering and X-ray self-absorption gives rise to the observed non-monotonic dependence. Notably, the right-hand y-axis (corresponding to the T_{accum}) closely reproduce the same trend seen in the left-hand y-axis (representing the total bremsstrahlung flux), explaining the non-monotonic increase in bremsstrahlung flux with crystal thickness.



Supplementary Fig. 12. **a** The left y-axis (blue) is bremsstrahlung flux from bulk MoSe₂ crystal with 2 mm thickness at 300 keV electron energy, which is the same as main text Fig 3b. The right y-axis (orange) is the accumulated transmission T_{accum} , as demonstrate in the Supplementary Equation (26), which also exhibit the same four regions. **b** The same analysis as **a** but for WSe₂.

S14. General rule of the ratio between tunable X-ray and bremsstrahlung

The framework proposed in the present work, which includes the electron scattering and tunable X-ray theory can accurately determine the tunable X-ray intensity. When combined with Monte Carlo simulations of bremsstrahlung radiation, it allows reliable prediction of the tunable X-ray to bremsstrahlung ratio across different materials, electron energies, and crystal orientations. To further elucidate the underlying physics, we identify a governing parameter and equation that capture the trends of this ratio as functions of material properties, emitted photons, and incident electrons.

In Figs. 1e and 1f of the main text, the x-axis corresponds to the first key parameter: the electron mean free path. Since tunable X-ray emission originates from electron trajectories inside the crystal, a longer mean free path increases the interaction time t_L in Supplementary Equation (2), thereby enhancing the tunable X-ray intensity. The y-axis corresponds to the governing equation, which incorporates material, photon, and electron properties, and can be expressed as containing two components,

$$\frac{E_p[\text{eV}]|\chi_g|^2}{\left(\frac{Z_{\text{eff}}}{E_p[\text{eV}]}\right) \times T_{\text{accum}}} = \frac{\text{Tunable X ray contribution}}{\text{Bremsstrahlung contribution}}, \quad (27)$$

Both components of the equation can be derived from the fundamental tunable X-ray and bremsstrahlung theory. The numerator represents the tunable X-ray contribution, and the term $E_p|\chi_g|^2$ can be derived from the fundamental equations of tunable X-ray theory. From Supplementary Equations (2) and (3) of Section S1 we get

$$\Phi_{\text{tunx}} \propto \omega|\mathbf{E}_{\mathbf{k}s}|^2 \propto \omega|\chi_g|^2 \propto E_p|\chi_g|^2, \quad (28)$$

where Φ_{tunx} is the tunable X-ray flux, ω is the X-ray frequency, $\mathbf{E}_{\mathbf{k}s}$ is the eigenmode of the crystal, χ_g is the electric susceptibility and E_p is photon energy.

The denominator of Supplementary Equation (27) corresponds to the bremsstrahlung contribution, the fundamental equation of bremsstrahlung intensity can be expressed as²⁹

$$\Phi_{\text{brem}} \propto n \frac{d\sigma}{dE_p} N_e e^{-\alpha L}, \quad (29)$$

where Φ_{brem} is the bremsstrahlung flux, n is the number of atoms per volume, $d\sigma/dE_p$ is the bremsstrahlung cross-section differential in emitted photon energy E_p , N_e is the electron distribution inside the crystal after electron scattering and $e^{-\alpha L}$ accounts for the X-ray self-absorption. The bremsstrahlung cross-section scales approximately as Z/E_p ³⁰, where Z is the atomic number and E_p is the X-ray photon energy. The combined effect of electron distribution after scattering and X-ray self-absorption can be represented by the accumulated transmission T_{accum} as

$$T_{\text{accum}} = \sum_{i=1}^{N_{\text{scatter}}} T_i = \sum_{i=1}^{N_{\text{scatter}}} e^{-\alpha L}, \quad (30)$$

which is equivalent to Supplementary Equation (26) mentioned before, and reproduces the bremsstrahlung dependence on crystal thickness. Substituting this into the Supplementary Equation (29) get

$$\Phi_{\text{brem}} \propto n \frac{d\sigma}{dE_p} N_e e^{(-\alpha L)} \propto n \left(\frac{Z}{E_p} \right) \times T_{\text{accum}} \propto \left(\frac{Z_{\text{eff}}}{E_p} \right) \times T_{\text{accum}}, \quad (31)$$

where $Z_{\text{eff}} = nZ$ is the effective atomic number of the crystal. Combining Supplementary Equations (28) and (31) we can get

$$\frac{\Phi_{\text{tunx}}}{\Phi_{\text{brem}}} \propto \frac{E_p[\text{eV}] |\chi_g|^2}{\left(\frac{Z_{\text{eff}}}{E_p[\text{eV}]} \right) \times T_{\text{accum}}}, \quad (32)$$

which defines the y-axis of main text Figs. 1f and 1e, where $E_p[\text{eV}] \equiv E_p/(1 \text{ eV})$ is the value of the photon energy normalized to one eV. The boundary equation (dotted purple line) in these figures marks the condition where tunable X-ray and bremsstrahlung fluxes are approximately equal. This boundary was obtained by fitting simulation data

points with ratios close to unity ($0.95 < \text{ratio} < 1.05$) and follows a power-law dependence:

$$\frac{E_p[\text{eV}]|\chi_g|^2}{\left(\frac{Z_{\text{eff}}}{E_p[\text{eV}]}\right) \times T_{\text{accum}}} = 9 \times 10^{-4} \times (\text{mean free path})^{-1.3}, \quad (33)$$

where the left-hand side corresponds to the governing equation we derived in Supplementary Equations (32), and the right-hand side depends on the electron mean free path, expressed in nanometers.

All data points shown in Figs. 1e and 1f are obtained using the theoretical framework and Monte Carlo simulation method introduced in this work for non-tilted crystals within the tabletop electron energy range of 10 – 300 keV. All simulations correspond to a solid angle of 0.066 steradians, and observation angle $\theta_{\text{obs}} = 119^\circ$ which is in accordance with experimental conditions. A variety of materials and crystal planes were considered, for each crystal, we consider tunable X-rays generated from four dominant crystal planes, where the values of $|\chi_g|$ are maximized. The contributions from all other crystal planes with less $|\chi_g|$ values are small compared to dominant planes (generally $< 30\%$), thus not considered in our simulation. Careful validation was performed to ensure the applicability of the classical theory, i.e., the quantum recoil is small in photon energies ($< 5\%$). The complete list of materials and crystal planes is provided in Supplementary Table 5.

Supplementary Table 5. List of simulated materials and crystal planes in main text Figs. 2a and 2b.

Material	Planes	Material	Planes	Material	Planes
Silicon	(111) (022) (113) (004)	SiO ₂	(101) (010) (011) (012)	MnPS ₃	(010) (130) (20-1) (11-1)
Diamond	(111) (022) (113) (004)	Graphite	(002) (004) (011) (010)	ZrS ₃	(10-1) (101) (10-2) (012)
LiF	(111) (002) (022) (113)	h-BN	(002) (011) (010) (004)	MnNb ₃ S ₆	(001) (002) (100) (1-10)
W	(011) (002) (112) (022)	MoSe ₂	(002) (013) (004) (010)	Nb ₃ FeS ₆	(002) (001) (110) (2-10)
Ge	(111) (022) (113) (004)	WSe ₂	(002) (004) (013) (011)	Nb ₃ NiSe ₆	(002) (001) (110) (004)
Cu	(111) (002) (022) (113)	TiS ₂	(003) (013) (100) (103)	CrPS ₄	(100) (200) (010) (201)
GaAs	(111) (002) (022) (113)	CrPS ₄	(100) (200) (010) (201)	V ₂ O ₅	(010) (001) (01-1) (002)

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